# STA2002 Tutorial 11

1. For the simple linear model $Y_i = \beta_1 x_i + \beta_0$, we proved that $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$. Please show that $\text{Var}(\hat{\beta}_0) = \frac{\overline{x^2}}{S_{xx}}\sigma^2$, where $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2$.

**Solution:**

$$\begin{aligned}\text{Var}\left(\hat{\beta}_0\right) &= \text{Var}\left(\bar{Y} - \hat{\beta}_1 \bar{x}\right) \\ &= \text{Var}(\bar{Y}) + (\bar{x})^2 \,\text{Var}\left(\hat{\beta}_1\right) - 2\bar{x}\,\text{Cov}\left(\bar{Y}, \hat{\beta}_1\right)\end{aligned}$$

We have

$$\text{Var}(\bar{Y}) = \text{Var}\left(\frac{1}{n}\sum_{i=1}^n Y_i\right) = \frac{1}{n^2}\sum_{i=1}^n \text{Var}\left(Y_i\right) = \frac{\sigma^2}{n}$$

and

$$\text{Var}\left(\hat{\beta}_1\right) = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{Sxx},$$

the covariance term is

$$\begin{aligned}\text{Cov}\left(\bar{Y}, \hat{\beta}_1\right) &= \text{Cov}\left\{\frac{1}{n}\sum_{i=1}^n Y_i, \frac{\sum_{j=1}^n (x_j - \bar{x})\, Y_j}{\sum_{i=1}^n (x_i - \bar{x})^2}\right\} \\ &= \frac{1}{n}\frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}\,\text{Cov}\left\{\sum_{i=1}^n Y_i, \sum_{j=1}^n (x_j - \bar{x})\, Y_j\right\} \\ &= \frac{1}{n\sum_{i=1}^n (x_i - \bar{x})^2}\sum_{i=1}^n (x_j - \bar{x})\sum_{j=1}^n \text{Cov}\left(Y_i, Y_j\right) \\ &= \frac{1}{n\sum_{i=1}^n (x_i - \bar{x})^2}\sum_{i=1}^n (x_j - \bar{x})\,\sigma^2 \\ &= 0 \quad (\text{as } \sum_{i=1}^n (x_i - \bar{x}) = 0)\end{aligned}$$

And since

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \sum_{i=1}^n x_i^2 - 2\bar{x}\sum_{i=1}^n x_i + \sum_{i=1}^n \bar{x}^2 = \sum_{i=1}^n x_i^2 - n\bar{x}^2,$$

we have

$$\begin{aligned}\text{Var}\left(\hat{\beta}_0\right) &= \frac{\sigma^2}{n} + \frac{\sigma^2\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sigma^2}{n\sum_{i=1}^n (x_i - \bar{x})^2}\left\{\sum_{i=1}^n (x_i - \bar{x})^2 + n\bar{x}^2\right\} \\ &= \frac{\sigma^2\sum_{i=1}^n x_i^2}{n\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\overline{x^2}}{S_{xx}}\sigma^2.\end{aligned}$$

2. Consider linear model $Y_i = aX_i + b + \varepsilon_i$, where $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$. Find the $(1-\alpha)\%$ confidence interval(CI) for $b$ under the case that $\sigma^2$ is known or unknown respectively. Also, what is the CI for $\sigma^2$?

**Solution:**

Since $\hat{b}$ is an unbiased estimator ($E(\hat{b}) = b$), and from the previous problem we know that $Var(\hat{b}) = \dfrac{\bar{X}^2\sigma^2}{S_{xx}}$. Then we have $\hat{b} = \bar{Y} - \hat{a}\bar{X} \sim N\left(b, \dfrac{\bar{X}^2\sigma^2}{S_{xx}}\right)$.

Let $\hat{\sigma^2} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{a}x_i - \hat{b})^2$, it can be shown that $n\hat{\sigma^2}/\sigma^2 \sim \chi^2(n-2)$.

- If $\sigma^2$ is known,

$$\hat{b} \pm z_{\alpha/2}\sqrt{\frac{\bar{X}^2}{S_{xx}}}\sigma$$

- If $\sigma^2$ is unknown,

$$\hat{b} \pm t_{\alpha/2}\sqrt{\frac{\bar{X}^2 n\hat{\sigma}^2}{(n-2)S_{xx}}}$$

Besides, the $(1-\alpha)\%$ CI for $\sigma^2$ is

$$\left[\frac{n\hat{\sigma^2}}{\chi^2_{\alpha/2}(n-2)}, \frac{n\hat{\sigma^2}}{\chi^2_{1-\alpha/2}(n-2)}\right]$$